

Karnatak University, Dharwad

QUESTION BANK

B.Sc.-V Semester (RNEP)(DSC)

Course Title: Real Analysis – II

Course Code: C5MAT1T1 (Theory)

Compulsory Theory paper

Unit-1: Riemann Integration-I:

Definition and examples for partition of an interval, Refinement of a partition and common refinement, Upper and lower Darboux sums, Upper and lower integrals, Riemann integral, Criterion for integrability, Integrability of sum, difference, product, quotient and modulus of integrable functions. Integral as a limit of sum (Riemann sum) –Problems.

Two marks:

1. Define norm of a partition.
2. Define refinement of a partition.
3. Define lower and upper Riemann sums.
4. If $f(x)$ is the function defined on $[a, b]$ by

$$f(x) = \begin{cases} 1, & x \text{ is rational} \\ -1, & x \text{ is irrational} \end{cases}$$

then find the oscillation of $f(x)$ in $[a, b]$

5. If $f(x) = x$ defined on $[0, 1]$ and $P = \left[0, \frac{1}{3}, \frac{2}{3}, 1\right]$ then find $U(P, f)$.
6. If $f(x) = x^2$ defined on $[0, 1]$ and $P = \left[0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1\right]$ then find $L(P, f)$.
7. If $f(x)$ is real valued bounded function defined on $[a, b]$ and $P \in P[a, b]$ then prove that $L(P, f) \leq U(P, f)$.
8. If $f(x)$ is real valued bounded function defined on $[a, b]$ and $P \in P[a, b]$, then prove that $L(P, -f) = -U(P, f)$.
9. If $f(x)$ is real valued bounded function defined on $[a, b]$ and $P \in P[a, b]$ then prove that $m(b-a) \leq L(P, -f) \leq U(P, f) \leq M(b-a)$.
10. If $f(x)$ is real valued bounded function defined on $[a, b]$ and $P \in P[a, b]$, then prove that $L(P, f) \leq L(P^*, f)$.
11. If $f(x)$ & $g(x)$ are real valued bounded functions defined on $[a, b]$ and $P \in P[a, b]$ then prove that $U(P, f + g) \leq U(P, f) + U(P, g)$.

12. If P_1 & P_2 are partition of then show that $L(P_1, f) \leq U(P_2, f)$.
13. Define Upper and lower Riemann Integrals.
14. State necessary and sufficient condition for Riemann integration.
15. Show that a constant function defined on $[a, b]$ is Riemann integrable.

Five marks:

1. If $f(x) = x^2$ defined on $[0, 1]$ and $P = \left[0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1\right]$ then find $U(P, f)$ & $L(P, f)$.
2. If $f(x)$ & $g(x)$ are real valued bounded functions defined on $[a, b]$ and $P \in P[a, b]$ then
 - (a) $U(P, f + g) \leq U(P, f) + U(P, g)$
 - (b) $L(P, f + g) \geq L(P, f) + L(P, g)$.
3. If $f(x)$ is bounded function defined on $[a, b]$ then $\int_{-a}^b f(x) dx \leq \int_a^{-b} f(x) dx$
4. If $f(x)$ bounded function on $[a, b]$, and M, m are respectively the supremum and infimum of $f(x)$ on $[a, b]$ then

$$m(b - a) \leq \int_{-a}^b f(x) dx \leq \int_a^{-b} f(x) dx \leq M(b - a).$$
5. If $f(x) \in R[a, b]$ and M, m are respectively the supremum and infimum of $f(x)$ on $[a, b]$ then $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$.
6. If $f(x) \in R[a, b]$ and M, m are respectively the supremum and infimum of $f(x)$ on $[a, b]$ then
 - (i) $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$ if $b \geq a$
 - (ii) $m(b - a) \geq \int_a^b f(x) dx \geq M(b - a)$ if $b \leq a$
7. State and prove necessary and sufficient condition for Riemann integration.
8. Show that $f(x) = x$ defined on $[0, 1]$ is R-integrable and $\int_0^1 f(x) dx = \frac{1}{2}$.
9. Prove that $f(x) = x^2 + 1$ defined on $[1, 2]$ is R-integrable and find $\int_1^2 f(x) dx$

10. Prove that the $f(x) = x^2 + x$ is defined on $[2, 3]$ is R-integrable and find

$$\int_2^3 f(x) dx.$$

11. Show that $f(x) = x^3$ defined on $[0, a]$ is R-integrable

$$\text{and } \int_0^a f(x) dx = \frac{a^4}{4}.$$

12. Show that $f(x)$ defined by $f(x) = \begin{cases} 1, & x \in Q \\ 0, & x \in Q' \text{ (or } R - Q) \end{cases}$ is not R-integrable over $[0, 1]$.

13. Show that $f(x)$ defined by $f(x) = \begin{cases} 1, & x \in Q \\ -1, & x \in Q' \text{ (or } R - Q) \end{cases}$ is not R-integrable over $[a, b]$.

14. If $f(x)$ is bounded & integrable function defined on $[a, b]$ and $\forall \epsilon > 0, \exists \delta > 0$

such that for every partition $P = [a = x_0, x_1, x_2, \dots, x_n = b]$ with $\|P\| < \delta$, choose $\xi \in [x_{r-1}, x_r]$ then

$$\left| \sum_{r=1}^n f(\xi_r) \cdot \delta_r - \int_a^b f(x) dx \right| < \epsilon$$

15. Show that $f(x) = 3x + 1$ defined on $[1, 2]$ is R-integrable

$$\text{and } \int_1^2 f(x) dx = \frac{11}{2}.$$

16. If $f(x) \in R[a, b]$ then $|f(x)| \in R[a, b]$ and $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$

17. If $f(x), g(x) \in R[a, b]$ & k be any scalar, then prove that

$$f(x) + g(x) \in R[a, b].$$

18. If $f(x), g(x) \in R[a, b]$ & k be any scalar, then prove that

$$f(x) \times g(x) \in R[a, b].$$

19. If $f(x), g(x) \in R[a, b]$ & k be any scalar, then prove that

$$\frac{f(x)}{g(x)} \in R[a, b], \quad g(x) \neq 0.$$

20. If $f(x) \in R[a, c]$ & $f(x) \in R[c, b]$, then $f(x) \in R[a, b]$.

Unit-2: Riemann Integration-II:

Integrability of continuous functions, Monotonic functions, Bounded functions with finite number of discontinuity, Fundamental theorem of Calculus – related problems, Change of variables, Integration by parts, First and second mean value theorems of integral calculus.

Two marks:

1. If $f(x)$ is continuous function defined on $[a, b]$ then there exists and $c \in (a, b)$ such that $\int_a^b f(x) dx = f(c)(b - a)$.
2. If $f(x)$ is monotonic function defined on $[a, b]$ then there exists and $c \in [a, b]$ such that $\int_a^b f(x) dx = (c - a)f(a) + (b - c)f(b)$.
3. State first mean value theorem for Integral Calculus.
4. State second mean value theorem for Integral Calculus.
5. State Bonnets form of mean value theorem.
6. Show that $\exists \mu \in \left[0, \frac{\pi}{2}\right]$ such that $\int_0^{\pi/2} x \sin x dx = \mu$.
7. Define primitive of a function and give an example.
8. State fundamental theorem of integral calculus.
9. Show that $\int_a^b \cos x dx = \sin b - \sin a$ using fundamental theorem of integral calculus.
10. Show that $\int_1^2 x^2 dx = \frac{7}{3}$ using fundamental theorem of integral calculus.
11. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using fundamental theorem of integral calculus.
12. State integration by change of variables.
13. If $f(x) \in R[0, a]$ then show that $\int_0^a f(x) dx = \int_0^a f(a - x) dx$.
14. State integration by parts.
15. Show that $\int_0^{\pi/2} x \cos x dx = \frac{\pi}{2} - 1$.

Five marks:

1. If $f(x)$ is continuous function on $[a, b]$ then prove that it is R-integrable on $[a, b]$.
2. If $f(x)$ is monotonic function on $[a, b]$ then prove that it is R-integrable on $[a, b]$.

3. A bounded function $f(x)$ on $[a, b]$, having a finite number of points of discontinuity in $[a, b]$ then prove that it is R-integrable on $[a, b]$.
4. State and prove first mean value theorem for integral calculus.
5. Show that $\frac{\pi}{4} \leq \int_0^{\pi/4} \sec x \, dx \leq \frac{\pi}{2\sqrt{2}}$ using mean value theorem.
6. Show that $\frac{3}{2\sqrt{2}} < \int_0^1 \frac{x^2}{\sqrt{1+x^2}} \, dx < \frac{1}{3}$ using first mean value theorem for integral calculus.
7. Show that $\int_0^{\pi} \frac{6x^2}{1+\cos x} \, dx > \pi^3$ using first mean value theorem for integral calculus.
8. Show that $\frac{\pi}{6} > \int_0^1 \frac{dx}{\sqrt{4-x^2+x^3}} > \frac{1}{2}$.
9. State and prove second mean value theorem for integral calculus.
10. Show that $\left| \int_1^2 \left(\frac{\sin x}{x} \right) dx \right| \leq 2$ using second mean value theorem for integral calculus.
11. If $f(x)$ is continuous on $[a, b]$ and if $\phi(x) = \int_a^x f(t) \, dt, \forall x \in [a, b]$ then prove that $\phi'(x) = f(x), \forall x \in [a, b]$.
12. State and prove fundamental theorem of integral calculus.
13. Evaluate $\int_0^{\pi/4} (\sec^4 x - \tan^4 x) \, dx$ using fundamental theorem of integral calculus.
14. If $f(x)$ is differentiable on $[a, b], f'(x) \in R[a, b], f(x) \neq 0, \forall x \in [a, b]$ then show that $\int_a^b \frac{f'(x)}{f(x)} \, dx = \log \left| \frac{f(b)}{f(a)} \right|$.
15. Using the substitution $x = \pi - t$ then show that $\int_0^{\pi} x \phi(\sin x) \, dx = \frac{\pi}{2} \int_0^{\pi} \phi(\sin x) \, dx$.
16. If $f(x)$ & $g(x)$ are differentiable functions on $[a, b]$ and $f'(x)$ & $g'(x)$ are continuous on $[a, b]$ then prove that $\int_a^b f(x) g'(x) \, dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x) g(x) \, dx$.

17. If $f(x)$ is differentiable function on $[a, b]$ and $f'(x)$ is continuous on $[a, b]$

then show that $\int_a^b [f(x) + f'(x)] e^x dx = [f(x) e^x]_a^b$.

18. Evaluate $\int_0^1 \left(\frac{1+x}{(2+x)^2} \right) e^x dx$.

Unit-3: Improper Integrals:

Improper integrals of the first, second and third kind with examples, Improper integral as the limit of proper integral, Comparison test Abel's test and Dirichlet's test for the convergence of the integral of a product of two functions.

Two marks:

1. Define improper integral of first kind and give an example.
2. Define improper integral of second kind and give an example.
3. Define improper integral of third kind and give an example.
4. Examine the convergence of the improper integrals;

$$(a) \int_1^{\infty} \frac{dx}{x} \quad (b) \int_0^{\infty} \frac{dx}{1+x^2} \quad (c) \int_0^{\infty} e^{-mx} dx \quad (m > 0)$$

5. Examine the convergence of the improper integrals;

$$(a) \int_0^1 \frac{dx}{x^2} \quad (b) \int_1^2 \frac{x}{\sqrt{x}-1} dx \quad (c) \int_0^{\frac{1}{e}} \frac{dx}{x(\log x)^2}$$

6. State test for convergence of $\int_a^b f(x) dx$ at a ,

7. Examine the convergence of the improper integrals:

$$(a) \int_0^1 \frac{dx}{\sqrt{x^2+x}} \quad (b) \int_0^1 \frac{dx}{\sqrt{1-x^3}} \quad (c) \int_2^3 \frac{x^2+1}{x^2-4} dx \quad (d) \int_0^{\pi/4} \frac{dx}{\sqrt{\tan x}}$$

8. State test for convergence of $\int_a^{\infty} f(x) dx$ at ∞ .

9. State Abel's test for convergence of the integral of a product of two functions.

10. State Dirichlet's test for convergence of the integral of a product of two functions.

Five marks:

1. Examine the convergence of the improper integrals;

$$(a) \int_1^{\infty} \frac{x}{(1+x)^3} dx \quad (b) \int_{-\infty}^{\infty} \frac{dx}{x^2+2x+2}$$

2. Examine the convergence of the improper integrals;

$$(a) \int_0^2 \frac{dx}{\sqrt{4-x^2}} \quad (b) \int_0^1 \frac{dx}{x^2-3x+2}$$

3. If $f(x)$ & $g(x)$ are two positive functions with $f(x) \leq g(x)$, $\forall x \in [a, b]$ and a is the only point of infinite discontinuity on $[a, b]$ then show that

$$(a) \int_a^b g(x) dx \text{ is convergent} \Rightarrow \int_a^b f(x) dx \text{ is convergent}$$

$$(b) \int_a^b f(x) dx \text{ is divergent} \Rightarrow \int_a^b g(x) dx \text{ is divergent}$$

4. If (a) $f(x) > 0$ & $g(x) > 0$ on $[a, b]$

$$(b) f(x) \text{ & } g(x) \text{ are continuous on } [a, b] \text{ &}$$

$$(c) \lim_{x \rightarrow a+} \frac{f(x)}{g(x)} = l \neq 0, \text{ where } l \text{ is some finite positive number then}$$

prove that $\int_a^b f(x) dx$ & $\int_a^b g(x) dx$ are either both converges or diverges..

5. If $f(x)$ & $g(x)$ are two positive functions on $[a, b]$, a being the only point of infinite discontinuity then prove that

$$(i) \lim_{x \rightarrow a+} \left[\frac{f(x)}{g(x)} \right] = 0 \text{ & } \int_a^b g(x) dx \text{ is converges} \Rightarrow \int_a^b f(x) dx \text{ is converges}$$

$$(ii) \lim_{x \rightarrow a+} \left[\frac{f(x)}{g(x)} \right] = \infty \text{ & } \int_a^b g(x) dx \text{ is diverges} \Rightarrow \int_a^b f(x) dx \text{ is diverges.}$$

6. Prove that the improper integral $\int_a^b \frac{dx}{(x-a)^n}$ (Improper integral of first kind) is convergent if $n < 1$.

7. Prove that the improper integral $\int_a^b \frac{dx}{(b-x)^n}$ (Improper integral of first kind) is convergent if $n < 1$.

8. Examine the convergence of the improper integrals:

$$(a) \int_0^1 \frac{dx}{\sqrt{x^2+x}} \quad (b) \int_0^1 \frac{dx}{\sqrt{1-x^3}} \quad (c) \int_2^3 \frac{x^2+1}{x^2-4} dx \quad (d) \int_0^{\pi/4} \frac{dx}{\sqrt{\tan x}}.$$

9. Discuss the convergence of Beta function i.e. $\int_0^1 x^{m-1} (1-x)^{n-1} dx$; $m, n > 0$.

10. Prove that the improper integral $\int_a^\infty \frac{dx}{x^n}$ (Improper integral of first kind) is convergent if $n > 1$.

11. Examine the convergence of the improper integrals:

$$(a) \int_0^{\infty} \frac{x}{x^3 + 1} dx \quad (b) \int_1^{\infty} \frac{x^3 + 1}{x^4} dx .$$

12. Discuss the convergence of Gamma function i.e. $\int_0^{\infty} e^{-x} x^{n-1} dx$.

13. State and prove Abel's test for convergence of the integral of a product of two functions.

14. Examine the convergence of integrals:

$$(a) \int_0^{\infty} e^{-ax} \frac{\sin x}{x} dx, \quad a \geq 0 \quad (b) \int_a^{\infty} e^{-x} \frac{\sin x}{x^2} dx, \quad a > 0 .$$

15. State and prove Dirichlet's test for convergence of the integral of a product of two functions.

16. Examine the convergence of integrals:

$$(a) \int_0^{\infty} \frac{\sin x}{x^{3/2}} dx \quad (b) \int_0^{\infty} \frac{x}{1 + x^2} \sin x dx$$

Unit-4: Beta-Gamma Functions:

Definitions, Properties and examples, Relations between Beta and Gamma functions, Standard theorems, Duplication formula and related problems.

Two marks:

1. Prove that $\Gamma(n+1) = n\Gamma(n)$.
2. Show that $\Gamma(n) = (n-1)!$.
3. Show that $\Gamma(n) = 2 \int_0^{\infty} e^{-x^2} x^{2n-1} dx$.
4. Show that $\Gamma(n) = t^n \int_0^{\infty} e^{-tx} x^{n-1} dx$.
5. Evaluate $\int_0^{\infty} e^{-x^{1/n}} dx = n\Gamma(n)$.
6. Prove that $\Gamma(n) = \int_0^1 \left(\log \frac{1}{y} \right)^{n-1} dy$.
7. Prove that $\int_0^{\infty} e^{-x^4} dx = \frac{1}{4} \Gamma\left(\frac{1}{4}\right)$.
8. Evaluate $\int_0^{\infty} \sqrt{y} e^{-y^2} dy$.
9. Show that $\beta\left(\frac{1}{2}, \frac{1}{2}\right) = \pi$.
10. Prove that $\int_{-1}^1 \sqrt{\frac{1+x}{1-x}} dx = \pi$.

11. Evaluate $\int_0^1 \frac{35x^3}{32\sqrt{1-x}} dx$.
12. Evaluate $\int_{-1}^1 \sqrt{\frac{1+x}{1-x}} dx = \pi$
13. Show that $\left| \frac{1}{4} \right| \left| \frac{3}{4} \right| = \sqrt{2} \pi$.
14. Show that $\beta\left(m, \frac{1}{2}\right) = 2^{2m-1} \beta(m, m)$.
15. Prove that $\beta(m, n) = \beta(m+1, n) + \beta(m, n+1)$.

Five marks:

1. Show that $\left| \frac{1}{2} \right| = \sqrt{\pi}$.
2. Prove that $\int_0^\infty \frac{x^c}{c^x} dx = \frac{\sqrt{c+1}}{(\log_e c)^{c+1}}$.
3. Prove that $\int_0^1 x^m \left(\log \frac{1}{x} \right)^n dx = \frac{\sqrt{n+1}}{(m+1)^{m+1}}$.
4. Show that $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$.
5. Show that $\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$.
6. Prove that $\beta(m, n) = 2 \int_0^\infty \frac{x^{2m-1}}{(1+x^2)^{m+n}} dx$.
7. Show that $\int_0^{\frac{\pi}{2}} \sin^p x \cos^q x dx = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$.
8. Derive the relation between Beta and Gamma functions i.e.

$$\beta(m, n) = \frac{\sqrt{m} \sqrt{n}}{\sqrt{m+n}}, m, n > 0.$$
9. Show that $\int_0^{\frac{\pi}{2}} \sin^p x dx \int_0^{\frac{\pi}{2}} \sin^{p+1} x dx = \frac{\pi}{2(p+1)}$.
10. Show that $\int_0^1 \left\{ \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} \right\} dx = \beta(m, n)$.
11. Show that $\int_{-1}^1 \frac{(1+x)^{2m-1} (1-x)^{2n-1}}{(1+x^2)^{m+n}} dx = 2^{m+n-2} \beta(m, n)$.

12. Prove that $\int_0^1 \frac{x^{m-1} (1-x)^{n-1}}{(a+x)^{m+n}} dx = \frac{\beta(m, n)}{a^n (1+a)^n}.$

13. Derive the Duplication formula for Gamma function i.e.

$$\Gamma(n) \Gamma\left(n + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2n-1}} \Gamma(2n).$$

14. $\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta = \frac{\pi}{\sqrt{2}}.$

15. Prove that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}.$

16. Prove that $\int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin \theta}} d\theta = \pi.$

17. Show that $\int_0^{\infty} \sqrt{y} e^{-y^2} dy \int_0^{\infty} \frac{e^{-y^2}}{\sqrt{y}} dy = \frac{\pi}{2\sqrt{2}}.$

18. Show that $\int_0^{\infty} x e^{-x^8} dx \int_0^{\infty} x^2 e^{-x^4} dx = \frac{\pi}{16\sqrt{2}}.$

Karnatak University Dharwad

Question Bank

BSc-V Semester

(RNEP 2026-27 & 2027-28)

DSE-1 (T6) Algebra-III and Special Functions (C5 MAT2 T1)

Unit- I: Rings – I, Integral Domain, Fields

2 Marks Questions

1. Define Ring.
2. Define Abelian Ring (Commutative ring) and give an example.
3. Define Ring with unity and give an example for commutative ring without unity.
4. Define Ring with zero divisors. Give an example.
5. Define Ring without zero divisors. Give an example.
6. State cancellation laws in Ring.
7. Define Units (invertible elements) of a ring with unity. Give an example.
8. Define integral domain, Give an example of a ring which is not an integral domain.
9. Define Field. Give an example of an integral domain which is not a field.
10. Define division ring (Skew field). Give an example of division ring which is not a field.
11. Prove that if $(R, +, \cdot)$ is a ring, then for all $a, b, c \in R$
 - i) $a \cdot 0 = 0 \cdot a = 0$
 - ii) $a \cdot (-b) = -(a \cdot b) = (-a) \cdot b$
 - iii) $(-a) \cdot (-b) = a \cdot b$
 - iv) $a \cdot (b - c) = a \cdot b - a \cdot c$
 - v) $(b - c) \cdot a = b \cdot a - c \cdot a$ (any one for two marks)
12. If $(R, +, \cdot)$ is a ring such that $a^2 = a, \forall a \in R$ then prove that $a + a = 0, \forall a \in R$

5 Marks Questions

1. Prove that the set $\mathbb{Z}$ of all integers is a ring with respect to addition and multiplication as two ring composition.
2. Prove that the set $2\mathbb{Z}$ of all even integers is a commutative ring without unity, the addition and multiplication of integers being the two ring composition.
3. Prove that the set $\mathbb{Q}$ of all rational numbers is a commutative ring with unity, the addition and multiplication of rational numbers being the two ring composition.
4. Prove that the set $\mathbb{R}$ of all real numbers is a commutative ring with unity, the addition and multiplication of real numbers being the two ring composition.
5. Prove that the set $\mathbb{C}$ of all complex numbers of the form $a + ib$, a, b are real number is a commutative ring with unity, the addition and multiplication of complex number being the two ring composition.
6. Prove that the set $\mathbb{R}$ of numbers of the form $a + b\sqrt{2}$, where a and b are integers is a ring with respect to usual addition and multiplication of numbers.

7. Show that the set of all matrices of the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $a, b, c, d \in \mathbb{Z}$, the set of all integers is a ring with respect to matrix addition and matrix multiplication. Is it a commutative ring with unity.
8. Show that the set of all matrices of the form, $\begin{pmatrix} 0 & x \\ 0 & y \end{pmatrix}$, $x, y \in \mathbb{Q}$ is a non – commutative ring without unity, the binary operations being matrix – addition and matrix – multiplication.
9. Prove that the set $R = \{0, 1, 2, 3, 4, 5\}$ is a commutative ring with respect to ' $+_6$ ' and ' $\times_6$ ' as the two ring compositions.
10. Prove that a ring R is without zero divisors if and only if the cancellation laws holds in R .
11. Prove that every field is an integral domain. Is the converse true? Justify your answer.
12. Prove that a division ring (Skew field) has no divisors of zero.
13. Prove that every finite integral domain is a field.

Unit -II: Rings – II, Ideals

2 Marks Questions

1. Define Subring and give an example
2. Define left ideal and right ideal of ring.
3. Prove that the set of $S = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$, $\forall a, b, c \in \mathbb{Z}$ is a subring of the ring $M_2(\mathbb{Z})$ of all 2×2 matrices over the set of integers.
4. Prove that the set of $S = \begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$, $\forall a, b \in \mathbb{Z}$ is a subring of the ring $M_2(\mathbb{Z})$ of all 2×2 matrices over the set of integers.
5. Define polynomial over an arbitrary ring R .
6. Define set of all polynomial over a ring R .
7. Define zero polynomial over ring R .
8. Define degree of polynomial over ring R .
9. State degree of the sum and product of two polynomials.
10. State and prove Factor theorem
11. State Remainder theorem
12. Find the polynomial $q(x)$ and $r(x)$ if $f(x) = x^5 - x^3 + 3x - 5$ and $g(x) = x^2 + 7$
13. Find the polynomial $q(x)$ and $r(x)$ if $f(x) = x^4 + 3x^3 + x^2 + 6x + 10$ and $g(x) = x^2 + x + 1$
14. If R is an integral domain with unity element, then prove that any unit in $R[x]$ must already be a unit in R .
15. Prove that the polynomial domain $F[x]$ over a field F is not a field.
16. Define greatest common divisor (gcd) of two polynomials over a field.
17. Define Prime or Irreducible polynomial over a field.
18. If p is a prime number, prove that the polynomial $x^n - p$ is irreducible over the field of rational numbers.
19. Show that the polynomial $x^2 - 3$ is irreducible over the field of rational numbers.
20. Define ideal of ring. Give an example.
21. Give an example to show union of two Ideals of a ring R need not be an Ideal.
22. Prove that every Left Ideal/Right Ideal/ Ideal of a ring R is a subring of Ring R .
23. If I is an ideal of a ring R with unity and $1 \in I$ prove that $I = R$.

24. Prove that a field has no proper ideals. OR If F is a field then its only ideal are (0) and F itself.
25. Prove that the set of real numbers of the form $\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}, \forall a, b \in \mathbb{Z}$ is a right ideal of the ring R of all real square matrices of order 2.
26. Prove that the set of real numbers of the form of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}, \forall a, b \in \mathbb{Z}$ is a left ideal of the ring R of all real square matrices of order 2.
27. Define Prime and Maximal Ideal

5 Marks Questions

1. Prove that a nonempty subset S of a ring R is a subring of R if and only if $\forall a, b \in S \Rightarrow a - b \in S$ and $a \cdot b \in S$
2. Prove that the intersection of two subring of ring R is a subring of R . Give an example to show union of two subrings need not be a subring.
3. Prove that the intersection of two ideals of ring R is an ideal of R .
4. Prove that a commutative ring with unity is a field if it has no proper ideals.
OR Let R be a commutative ring with unit element whose ideal are (0) and R itself. Then prove that R is a field.
5. Let R be a ring with unit element, such that only right ideals of R are (0) and R . Prove that R is a division ring.
6. Let R be a ring with unit element, such that only left ideals of R are (0) and R . Prove that R is a division ring.
7. If P is an integer then $p\mathbb{Z} = \{pn : n \in \mathbb{Z}\}$ is a maximal ideal of $\mathbb{Z}$ if and only if p is prime.
8. Find the degree of sum and product of the following polynomials over the ring of integers $f(x) = 2x^0 + 5x + 3x^2 - 4x^3, g(x) = 3x^0 + 4x - x^3 + 5x^4$
9. Find the degree sum and product of the following polynomials over the ring $(I_6, +_6, \times_6)$
 $f(x) = 2x^0 + 5x + 3x^2, g(x) = 1x^0 + 4x + 2x^3$
10. Find the degrees of $[f(x) + g(x)], [f(x) \cdot g(x)], [h(x) \cdot h(x) \cdot h(x)]$ where the polynomials are on the integer (mod 8) and, operations are addition and multiplication of
 $f(x) = 2x + 4x^2, g(x) = 2 + 6x + 4x^2, h(x) = 2 + 4x$
11. Prove that the set $R[x]$ of all polynomials over an arbitrary ring R is a ring with respect to addition and multiplication of polynomials.
12. If R is an integral domain, then prove that the polynomial ring $R[x]$ is also an integral domain.
13. If $f(x), g(x) \in F(x)$ with $g(x) \neq 0$, then prove that there exists unique elements $q(x)$ and $r(x)$ of $F(x)$ such that $f(x) = q(x)g(x) + r(x)$ where either $r(x) = 0$ or $\deg f(x) < \deg g(x)$
14. Find the g. c. d of the polynomials $f(x) = x^4 + x^3 + 2x^2 + x + 1$, and $g(x) = x^3 - 1$ over the field of R of rational numbers and express it as a linear combination of two polynomials.
15. Find the g. c. d of the polynomials $f(x) = x^3 + 2x^2 + 3x + 2$, and $g(x) = x^2 - 4$ over the field of residue class modulo 5 and express it as a linear combination of two polynomials.
16. State and prove unique factorization theorem.
17. State and prove Eisenstein's Criterion of Irreducibility.

Unit-III: Bessel's Functions

2 Marks Questions

1. Define ordinary point and regular singular point of a differential equation.
2. Show that $x = 0$ is an ordinary point of the following differential equations

i) $(x^2 + 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - xy = 0$

ii) $(x^2 - 1) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$

iii) $\frac{d^2y}{dx^2} - x \frac{dy}{dx} + 2y = 0$

3. Show that $x = 0$ is a regular singular point of the following differential equations

i) $2x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + (x + 1)y = 0$

ii) $x^2(x + 1)^2 \frac{d^2y}{dx^2} + (x^2 - 1) \frac{dy}{dx} + 2y = 0$

iii) $x(x - 1)^3 \frac{d^2y}{dx^2} + 2(x - 1)^2 \frac{dy}{dx} + 3y = 0$

iv) $2x^2 \frac{d^2y}{dx^2} + 7x(x + 1) \frac{dy}{dx} - 3y = 0$

4. Show that $x = 0$ is an irregular singular point of the differential equation

$$x^2(x - 1)^2 \frac{d^2y}{dx^2} + 3(x - 1) \frac{dy}{dx} + 7y = 0$$

5. Find the ordinary points, regular singular point and irregular singular point of the following differential equations

i) $2x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + (x + 1)y = 0$

ii) $x^3 \frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 6(x + 1)y = 0$

iii) $(x - 1)(x + 2) \frac{d^2y}{dx^2} + x^2 \frac{dy}{dx} + xy = 0$

iv) $(x - 1)^4 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$

6. Write the generating function for $J_n(x)$.
7. Write the value of $J_{\frac{1}{2}}(x)$ and $J_{-\frac{1}{2}}(x)$.
8. Show that $J'_0(x) = -J_1(x)$.
9. Define Bessel's differential equation of order n .
10. Write the relation between $J_n(x)$ and $J_{-n}(x)$ when n is any integer.

5 Marks Questions

1. Find the power series solution of the following differential equations

i) $\frac{dy}{dx} - 2xy = 0$

ii) $\frac{d^2y}{dx^2} - y = 0$

iii) $\frac{d^2y}{dx^2} + xy = 0$

iv) $\frac{d^2y}{dx^2} + y = 0$

v) $\frac{d^2y}{dx^2} + 2xy = 0$

vi) $\frac{d^2y}{dx^2} + x^2y = 0$

vii) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$

viii) $\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$

ix) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + x^2y = 0$

x) $(x-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$

2. Solve the following differential equations in series using Frobenius method

i) $4x\frac{d^2y}{dx^2} + y = 0$

ii) $9x\frac{d^2y}{dx^2} + y = 0$

iii) $x\frac{d^2y}{dx^2} + x^2y = 0$

iv) $16x\frac{d^2y}{dx^2} + y = 0$

v) $2x\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$

vi) $4x\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$

vii) $4x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$

viii) $9x\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 0$

ix) $2x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - (x+1)y = 0$

x) $9x(1-x)\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 4y = 0$

3. Solve the Bessel's differential equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0$

OR Obtain the solution for Bessel's differential equation $\frac{d^2y}{dx^2} + \frac{1}{x}\frac{dy}{dx} + \left(1 - \frac{n^2}{x^2}\right)y = 0$

4. Show that $J_{-n}(x) = (-1)^n J_n(x)$ where n is any integer.

5. Derive the generating function for $J_n(x)$.

OR Prove that $e^{\left\{\frac{x}{2}\left(t - \frac{1}{t}\right)\right\}} = \sum_{n=-\infty}^{\infty} t^n J_n(x)$.

6. Show that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ and $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$

7. Show that $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ and hence show that $xJ'_n(x) = xJ_{n-1}(x) - nJ_n(x)$

OR Show that $xJ'_n(x) = xJ_{n-1}(x) - nJ_n(x)$

8. Show that $\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$ and hence show that $xJ'_n(x) = nJ_n(x) - xJ_{n+1}(x)$

OR Show that $xJ'_n(x) = nJ_n(x) - xJ_{n+1}(x)$

9. Prove that the relation $2J'_n(x) = J_{n-1}(x) - J_{n+1}(x)$

10. i) Express $J_2(x)$ and $J_3(x)$ in terms of $J_0(x)$ and $J_1(x)$ using recurrence relations.

ii) Express $J_4(x)$ in terms of $J_0(x)$ and $J_1(x)$ using recurrence relations.

iii) Express $J_5(x)$ in terms of $J_0(x)$ and $J_1(x)$ using recurrence relations.

Unit -IV: Legendre's Functions

2 Marks Questions

1. Define Legendre's differential equation.
2. Write the generating function for $P_n(x)$.
3. State the orthogonality property of Legendre polynomials.
4. Write Rodrigue's formula for $P_n(x)$.
5. Find i) $P_0(x)$ ii) $P_1(x)$ iii) $P_2(x)$.

OR Show that i) $P_0(x) = 1$ ii) $P_1(x) = x$ iii) $P_2(x) = \frac{1}{2}(3x^2 - 1)$

6. Show that $P_n(1) = 1$
7. Show that $P_n(-1) = (-1)^n$.
8. Show that $\int_{-1}^1 P_0(x) dx = 2$
9. Show that $\int_{-1}^1 P_1(x) dx = 0$

5 Marks Questions

1. Solve Legendre's differential equation $(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$
2. Prove that Rodrigue's formula: $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$.
3. Obtain the generating function for Legendre polynomials $P_n(x)$.

OR Show that $(1 - 2xh + h^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} h^n P_n(x)$

4. Prove that the recurrence relation $(2n+1)x P_n(x) = (n+1)P_{n+1}(x) + n P_{n-1}(x)$.

OR Prove that the recurrence relation $(n+1)P_{n+1}(x) = (2n+1)x P_n(x) - n P_{n-1}(x)$.

5. Show that $(1 - x^2)P'_n(x) = n[P_{n-1}(x) - x P_n(x)]$.
6. Show that the recurrence relation $n P_n(x) = x P'_n(x) - P'_{n-1}(x)$.
7. Show that $P_n(-x) = (-1)^n P_n(x)$.
8. Show that $\int_{-1}^1 P_n(x) dx = 0$ if $n \neq 0$
9. Prove the orthogonality relation

i) $\int_{-1}^1 P_m(x) P_n(x) dx = 0$ if $m \neq n$,

ii) $\int_{-1}^1 (P_n(x))^2 dx = \frac{2}{2n+1}$ if $m = n$.

10. Express the function $f(x) = x^3 - 5x^2 + x + 1$ into Legendre polynomial.
11. Express the function $f(x) = 1 + 2x - x^2$ into Legendre polynomial.
12. Express the following function into Legendre polynomial $f(x) = x^3 + 2x^2 - x - 3$
13. Prove that $P'_{n+1}(x) + P'_{n-1}(x) = P_0(x) + 3P_1(x) + 5P_2(x) + \dots + (2n+1)P_n(x)$

OR Prove that $P'_{n+1} + P'_{n-1} = P_0 + 3P_1 + 5P_2 + \dots + (2n+1)P_n$

14. Prove that i) $P_{2m+1}(0) = 0$ and ii) $P_{2m}(0) = (-1)^n \frac{2m!}{2^{2m}(m!)^2}$

Karnatak University Mathematics Teacher's Association (KUMTA)

SPECIALIZATION – II

B.Sc. Semester – V (RVSD NEP) Question Bank

Subject: MATHEMATICS

Discipline Specific Core Elective (DSE) Course

Title: Linear Algebra and Analytical Geometry

Course Code: C5MAT2T2 (Theory)

Units	Linear Algebra and Analytical Geometry	60 Hours / Sem
Unit-I	Vector Space: Introduction to Rings, Fields and Vector space with examples, properties, Vector, Subspace. Linear independence and dependence of vectors. Linear span of a set. Base for vector space and dimension.	15 Hrs.
Unit-II	Linear transformation: Definition and examples, properties, range, null space, rank and nullity theorem, Linear maps as matrices-change of basic and effect of associated matrices, eigenvalues and eigenvectors of a linear transformation.	15 Hrs.
Unit-III	Sphere: Equation of a Sphere. Equation of a sphere with two given points as the ends of diameter (Different forms), Intersection of two spheres, Section of a sphere by a plane. Equation of a Sphere through a circle. Tangent planes. Orthogonal spheres.	15 Hrs.
Unit-IV	Cone and Cylinder: Equation of a cone, Quadric cone. Enveloping cone of a sphere, Right circular cone, Equation of a cylinder, Envelop	15 Hrs.

Unit I and Unit II

Linear Algebra and Linear transformation:

Questions carrying Two Marks each

- 1) Define vector space.
- 2) If any vector space V over a field F , prove any one of the following.
 - i) $\alpha 0 = 0 \quad \forall \alpha \in F$
 - ii) $0x = 0 \quad \forall x \in V$
 - iii) $(-\alpha)x = -(\alpha x) = \alpha(-x) \quad \forall x \in V \text{ and } \forall \alpha \in F$
 - iv) If $\alpha x = 0$ then either $\alpha = 0$ or $x = 0$
 - v) $\alpha x = \beta x$ then $\alpha = \beta$ if $x \neq 0$ for $x \in V, \alpha, \beta \in F$
 - vi) $\alpha x = \alpha y$ then $x = y$ if $\alpha \neq 0$ for $\alpha \in F, x, y \in V$.
- 3) Define a subspace of a vector space.
- 4) Show that a non-empty subset W of a vector space V over a field F is subspace of V if and only if $\alpha x + \beta y \in W \quad \forall x, y \in W \text{ and } \forall \alpha, \beta \in F$
- 5) Show that $W = \{(x, y, z) \mid x - 3y + 4z = 0, x, y, z \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3(\mathbb{R})$.
- 6) Show that the union of two subspaces of a vector space need not be a subspace.
- 7) If S and T are subsets of a vector space $V(F)$ then P.T
 - a) $S \subseteq L(T) \Rightarrow L(S) \subseteq L(T)$
 - b) $S \subseteq T \Rightarrow L(S) \subseteq L(T)$
 - c) S is a subspace of $V \Leftrightarrow L(S) = S$

d) $L[L(S)] = L(S)$

e) $L(S \cup T) = L(S) + L(T)$

f) $L[S]$ is the smallest subspace of V containing S .

8) Define Linearly dependent and Linearly independent vectors of a vector space.

9) Show that the vectors $x_1 = (1, 0, 0)$, $x_2 = (0, 1, 0)$ and $x_3 = (0, 0, 1)$ are linearly independent.

10) Show that the set of vectors containing zero vector '0' is linearly independent.

11) Show that the set consisting of a single vector α of V is linearly independent if and only if $\alpha \neq 0$.

12) Define the Linear span of a vector space.

13) Define dimension of finite dimensional vector space.

17) Define basis of a vector space.

18) Show any one of the following:

1) $\{(1,0,1), (0,2,2), (3,7,1)\}$ is a basis of $\mathbb{R}^3(\mathbb{R})$.

2) $\{(1,2,1), (3,4,-7), (3,1,5)\}$ is a basis of $\mathbb{R}^3(\mathbb{R})$.

3) $\{(1,2,1), (2,1,0), (1,-1,2)\}$ is a basis of $\mathbb{R}^3(\mathbb{R})$.

19) Define a Linear transformation.

20) If $T : V(F) \rightarrow V'(F)$ is a linear transformation then prove any two of the following:

i) $T(0) = 0'$ where 0 and $0'$ are the zero vectors of V and V' respectively.

ii) $T(-x) = -T(x) \forall x \in V$ and

iii) $T(x - y) = T(x) - T(y) \forall x, y \in V$

21) Define range and kernel (null space) of a linear transformation.

22) Define rank of a linear transformation.

23) Define nullity of a linear transformation.

Let $U(F)$ and $V(F)$ be two vector spaces. Then prove that

$N(T) = \{x: x \in U, T(x) = 0\}$ is a sub-space of $U(F)$.

26) Define the order basis of vector space.

27) Find the matrix of the linear transformation $T: V_3(\mathbb{R}) \rightarrow V_2(\mathbb{R})$

defined by $T(x, y, z) = (x+y, y+z)$ relative to the basis

$$B = \{(1, 1, 1), (1, 0, 0), (1, 1, 0)\} \text{ of } V_3(\mathbb{R})$$

28) Find the matrix of the linear transformation $T: V_3(\mathbb{R}) \rightarrow V_2(\mathbb{R})$

defined by $T(x, y, z) = (x+y, y+z)$ relative standard bases of $V_3(\mathbb{R})$ and $V_2(\mathbb{R})$.

29) Define the change of Basis.

30) P.T if V and W are finite dimensional vector space of the same dimension then a linear map $T: V \rightarrow W$ is one- one if and only if T is onto

31) Let $T: V \rightarrow W$ be a linear transformation. Then prove that T is non-singular if and only if $\text{rank of } T = \dim V$

32) Define the Eigen values and Eigen vectors of linear transformation.

34) Find the eigen values and corresponding eigen vectors of the identity linear transformation.

35) Show that 0 is an eigen value of T if and only if T is singular

36) If λ is an eigen value of an invertible operator T on a vector space V over F , then λ^{-1} is an eigen value of T^{-1} .

Questions carrying Five Marks each

- 1) Let $V(F)$ be a vector space and 0 be the Zero vector then P.T
 $a\alpha = 0 \Rightarrow \alpha = 0$ or $\alpha = 0 \quad \forall a \in F$ and $\forall \alpha \in V$
- 2) Prove that the set of all real valued continuous functions of x defined in intervals $[0,1]$ is a vector space.
- 3) Let $V(F)$ be vector space, $a, b \in F$ and $\alpha, \beta \in V$ be non zero elements then prove that
i) $a\alpha = b\alpha \Rightarrow a = b$
ii) $a\alpha = a\beta \Rightarrow \alpha = \beta$
- 4) Let $V(F)$ be a vector space and $a \in F$ and $\alpha, \beta \in V$ then prove that $a(\alpha - \beta) = a\alpha - a\beta$
- 5) Prove that a non-empty subset W of a vector space V over a field F is a subspace if and only if
 - (i) $x+y \in W \quad \forall x, y \in W$
 - (ii) $\alpha x \in W \quad \forall \alpha \in F$ and $\forall x \in W$.
- 6) Prove that intersection of any two subspaces w_1 and w_2 of a vector space $V(F)$ is also subspace of $V(F)$.
- 7) Show that union of two subspaces of a vector space $V(F)$ is a subspace if and only if one is contained in the other.

8) Show that $V_n(F) = \{(x_1, x_2, \dots, x_n) \mid x_i \in F\}$, Where F is a field is a vector space with addition and scalar multiplication defined by

$$(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) =$$

$$(x_1 + y_1, x_2 + y_2, \dots, x_n + y_n), \alpha (x_1, x_2, \dots, x_n) =$$

$$(\alpha x_1, \alpha x_2, \dots, \alpha x_n)$$

9) If A is any non-empty subset of a vector space $V(F)$, then show that $L[A]$ the linear span of A is the smallest subspace of V containing A .

10) For any non-empty subsets A, B of a vector space $V(F)$, prove

i) if $A \subset B$ then $L[A] \subset L[B]$ and

ii) $L[A] = A$ if and only if A is a subspace of V .

11) Show that $W = \{(x, y, z) \mid x - 3y + 4z = 0, x, y, z \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3(\mathbb{R})$.

12) Show that the following sets are sub-spaces of the vector space $\mathbb{R}^3(\mathbb{R})$ (any one)

i) $X = \{(x, 0, 0) \mid x \in \mathbb{R}\}$

ii) $X = \{(0, y, z) \mid y, z \in \mathbb{R}\}$

iii) $W = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathbb{R}\}$

13) Show that any subset of a linearly independent set of a vector space is also linearly independent.

14) Show that a superset of a linearly dependent set is also linearly dependent.

16) Prove that the set of vectors $(0, 2, -4)$, $(1, -2, -1)$ and $(1, -4, 3)$ of $V_3(\mathbb{R})$ are linearly dependent.

- 17) Prove that the set of vectors $(1, 3, 2)$, $(1, -7, -8)$ and $(2, 1, -1)$ of $V_3(\mathbb{R})$ are linearly dependent.
- 18) Express the vector $(2, -5, 3)$ as a linear combination of the vectors $(1, -3, 2)$, $(2, -4, -1)$ and $(1, -5, 7)$ in the vector space $\mathbb{R}^3(\mathbb{R})$
- 19) Prove that linear space $L(S)$ of any subset S of a vector space $V(F)$ is a subspace of V .
- 20) Show that $S = \{(1, 2, 4), (1, 0, 0), (0, 0, 1)\}$ is a linearly dependent subset of the vector space $V_3(\mathbb{R})$.
- 21) Show that any two bases of a finite dimensional vector space have the same number of elements.
- 22) Find the dimension and basis of the subspace spanned by the vectors $(2, 4, 2)$, $(1, -1, 0)$, $(1, 2, 1)$ and $(0, 2, 1)$ in $V_3(\mathbb{R})$.
- 24) If $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is any basis of a vector space $V(F)$ and $\beta_1, \beta_2, \dots, \beta_n$ are any n vectors of a vector space $W(F)$ then prove that there exists one and only one linear transformation $T: V \rightarrow W$ with $T(\alpha_i) = \beta_i$ for $i = 1, 2, \dots, n$.
- 25) If $T: V(F) \rightarrow W(F)$ is a linear transformation then show that $T(c_1 x_1 + c_2 x_2 + \dots + c_n x_n) = c_1 T(x_1) + c_2 T(x_2) + \dots + c_n T(x_n) \forall c_1, c_2, \dots, c_n \in F$ and $\forall x_1, x_2, \dots, x_n \in V$
- 26) Let $T: V(F) \rightarrow W(F)$ be a linear transformation. Then prove that
- range $R(T)$ is a subspace of W .
 - Kernel $N(T)$ is a subspace of V and
 - T is one-one if and only if $N(T) = \{0\}$

28) State and prove rank nullity theorem.

29) Show that the following mappings T are linear transformation

(answer any one)

i) $T: R^2 \rightarrow R^2$ defined by $T(x,y)=(x+y, x)$

ii) $T: R^2 \rightarrow R^3$ defined by $T(x,y)=(x+y, x-y, y)$

iii) $T: R^3 \rightarrow R^2$ defined by $T(x,y,z)=(3x-2y+z, x-3y-2z)$

iv) $T: R^2 \rightarrow R^2$ defined by $T(x,y)=(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$

v) $T: R^3 \rightarrow R^2$ defined by $T(x,y,z)=(x+y, y+z)$

30) Find the linear transformation $f: R^2 \rightarrow R^2$ such that

i) $f(1, 0) = (1, 1)$ and $f(0, 1) = (-1, 2)$.

ii) $f(1,1) = (0, 1)$ and $f(-1, 1) = (3, 2)$

iii) $f(2,3) = (4, 5)$ and $f(1, 0) = (0, 0)$ (Answer any one)

31) Find the matrix of the linear transformation $T: V_2(R) \rightarrow V_3(R)$

Defined by $T(x, y) = (x+y, x, 3x-y)$ with respect to standard bases.

32) Let $T: V \rightarrow W$ be a linear transformation defined by

i) $T(x, y, z) = (x + y, x - y, 2x + z)$.

ii) $T(x, y, z) = (y-x, y-z)$ (Answer any one)

Find the range, null space, rank, nullity and hence verify the rank – nullity theorem

33) Let V be a Vector space and T a linear transformation from V. Prove that the following two statements are equivalent:

i) The intersection of range of T and null space of T is the zero sub-space of T

ii) If $T(T\alpha) = 0$, then $T(\alpha) = 0$

34) Let $V(F)$ be an n -dimensional vector space and T a linear operator on V such that $\text{rank } T^2 = \text{rank } T$. Prove that the range and null space of T are disjoint.

35) Let U, V and W be three vector spaces over the field F . Let T_1, T_2 be linear transformations from U to V and S_1, S_2 be linear transformation from v to W then P.T

i) $S_1(T_1 + T_2) = S_1T_1 + S_2T_2$

ii) $(S_1 + S_2)T_1 = S_1T_1 + S_2T_1$

iii) $\alpha(S_1T_1) = (\alpha S_1)T_1 = S_1(\alpha T_1)$

.

38) Find the matrix representation of the following linear mappings relative to the usual basis of R^n .

$T: R^2 \rightarrow R^3$ defined by $T(x,y)=(3x-y, 2x+4y, 5x-6y)$

42) Prove that Non- zero vectors belonging to distinct values are linearly independent.

45) Find the eigen values and a basis of each eigen space of the linear operator $T: R^2 \rightarrow R^3$ defined by $T(x,y)=(3x+3y, x+5y)$.

49) Show that $T: R^3 \rightarrow R^2$ defined by $T(x,y,z)=(x-y, x-z)$ is a linear transformation

50) Show that $T: R^3 \rightarrow R$ defined by $T(x,y,z)=(2x-3y+4z)$ is a linear transformation

51) If $V(F)$ and $W(F)$ be two vector spaces and T_1 and T_2 are linear transformation from V into W prove that the mapping T defined by $T(\alpha)=c T_1(\alpha)+ T_2(\alpha), \alpha \in V, c \in F$ is a linear transformation.

52) Find the matrix representation of the operator T on $\mathbb{R}^2$ relative to the basis $B_1 = \{(1,3), (2,5)\}$; $T(x,y) = (2y, 3x-y)$

53) Find the matrix representation of the operator T on $\mathbb{R}^2$ relative to the basis $B_1 = \{(1,0), (0,1)\}$; $T(x,y) = (3x-4y, x+5y)$

54) Find out the matrix of a linear transformation

$T: V_2(\mathbb{R}) \rightarrow V_3(\mathbb{R})$ defined by $T(x,y) = (-x+2y, y, -3x+3y)$ relative to the basis $B_1 = \{(1,1), (-1,1)\}$ and $B_2 = \{(1,1,1), (1,-1,1), (0,0,1)\}$

i) $A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$

ii) $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$

Unit III

SPHERE

Questions carrying Two Marks each

1. Define a sphere.

2. Find the equation of a sphere with

i) center $(2,-3,4)$ and radius 5

ii) center $(-1,2,-3)$ radius 4

iii) center $(3,-4,1)$ radius 5

3. Find the centers and radii of the spheres

i) $x^2 + y^2 + z^2 - 6x + 8y - 10z + 1 = 0$

ii) $x^2 + y^2 + z^2 + 2x - 4y - 6z + 5 = 0$

iii) $2x^2 + 2y^2 + 2z^2 - 2x + 4y + 2z + 3 = 0$

iv) $x^2 + y^2 + z^2 - 2x + 4y - 6z = 2$

4. Find the equation of the sphere described on the join of points

as diameter.

- i) $A(2,-3,4)$ $B(-5,6,-7)$
- ii) $A(1,-1,3)$ $B(5,6,-1)$
- iii) $A(1,-2,-3)$ $B(1,-2,-1)$
- iv) $A(1,-2,-3)$ $B(-1,2,1)$

Questions carrying Five Marks each

5. Prove that the equation

$x^2 + y^2 + z^2 + 24x + 24y + 24z + 1 = 0$ represents a sphere and find its centre and radius.

6. Find the equation of the sphere through the four points

i) $(4,-1,2), (0,-2,3), (1,-5,-1), (2,0,1)$

ii) $(0,0,0), (0,1,-1), (-1,2,0), (1,2,3)$

iii) $(0,0,0), (-1,1,1), (1,-1,1), (1,1,-1)$

iv) $(3,0,2), (-1,1,1), (2,-5,4)$ and having its centre on the plane $2x + 3y + 4z = 6$

v) $(1,-3,4), (1,-5,2), (1,-3,0)$ and having its centre on the plane $x + y + z = 0$

4. Show that the spheres $x^2 + y^2 + z^2 = 25$ and $x^2 + y^2 + z^2 - 18x - 24y - 40z + 25 = 0$ touch and find the coordinates of their point of contact.

ii) $x^2 + y^2 + z^2 = 64$ and $x^2 + y^2 + z^2 - 12x + 4y - 6z + 48 = 0$ touch internally and find the point of contact

5. Prove that the plane section of a sphere is a circle and what is the centre and radius of that circle.

6. Find the centre and radius of a circle in which the sphere $x^2 + y^2 + z^2 + 2x - 2y - 4z - 19 = 0$ is cut by the plane $x + 2y + 2z + 7 = 0$.

7. Find the centre and radius of the circle $x^2 + y^2 + z^2 = 25$; $2x + y + 2z = 9$.
8. Find the centre and radius of the circle $x^2 + y^2 + z^2 + x + y + z - 4 = 0$; $x + y + z = 0$.
9. Find the centre and radius of the circle $x + 2y + 2z = 15$; $x^2 + y^2 + z^2 - 2y - 4z = 11$.
10. Find the equation of the sphere through the circle $x^2 + y^2 + z^2 = 9$; $2x + 3y + 4z = 5$ and the point $(1, 2, 3)$.
11. Find the equation of the sphere through the circle $x^2 + y^2 + z^2 + 2x + 3y + 6 = 0$; $x - 2y + 4z - 9 = 0$.
12. Show that the equation of the sphere having its centre on the plane $4x - 5y - z = 5$ and passing through the circle $x^2 + y^2 + z^2 - 2x - 3y + 4z + 8 = 0$; $x^2 + y^2 + z^2 + 4x + 5y - 6z - 28 = 0$; is $x^2 + y^2 + z^2 + 7x + 9y - 11z - 1 = 0$.
13. Define great circle.
14. Find the equation of the sphere for which the circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0$; $2x + 3y + 4z = 8$ is a great circle.
15. Find the sphere for which the circle $x^2 + y^2 + z^2 + 10y - 4z + 8 = 0$; $x + y + z = 3$ is a great circle.
16. Find the equation of the sphere through the circle $x^2 + y^2 + z^2 + 2x + 3y + 6 = 0$; and the centre of the sphere $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0$;
17. Obtain the equation of the sphere having circle $x^2 + y^2 + z^2 = 9$ $x - 2y + 2z = 5$ for a great circle and determine its center and radius.
18. Derive the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ at any point (x_1, y_1, z_1) on it.
19. Find the equation of the tangent plane to the sphere $3x^2 + 3y^2 + 3z^2 - 2x - 3y - 4z - 22 = 0$ at point $(1, 2, 3)$.

20. Show that the plane $2x-2y+z+12=0$ touches the sphere $x^2+y^2+z^2-2x-4y+2z-3=0$ and find the point of contact.
21. Show that the plane $2x+y-z=12$ touches the sphere $x^2+y^2+z^2=24$ and find the point of contact.
22. Show that the plane $2x-y-2z=4$ is a tangent plane to the sphere $x^2+y^2+z^2+2x-6y+1=0$ and also the point of contact.
23. Find the equations of the tangent planes to the sphere $x^2+y^2+z^2-4x+2y-6z+5=0$ and parallel to the plane $2x+2y-2=0$.
24. Find the equations of the tangent planes to the sphere $x^2+y^2+z^2-4x+2y-6z+5=0$ and parallel to the plane $2x+2y=5$
25. Define angle between two spheres
26. Derive the condition for the two spheres to intersect orthogonally.
27. Show that the spheres $x^2+y^2+z^2+6y+2z+8=0$ and $x^2+y^2+z^2+6x+8y+4z+20=0$ are orthogonal.
28. Show that spheres $x^2+y^2+z^2+2x+4y-6z+11=0$ and $x^2+y^2+z^2-x-y+8z+7=0$ cut orthogonally.
29. Show that spheres $x^2+y^2+z^2-4x+6y+4=0$ and $x^2+y^2+z^2+7x+10y-5z+12$ cut orthogonally.
30. Find k such that the spheres are orthogonal
 - i) $x^2+y^2+z^2+4x+ky+2z+6=0$ & $x^2+y^2+z^2+2x-4y-2z+6=0$
 - ii) $x^2+y^2+z^2+7x+ky-5z+12=0$ & $x^2+y^2+z^2-4x+6y+4=0$.

Unit IV

CONES AND CYLINDERS

**Questions carrying Two Marks each
and Questions carrying Five Marks each**

1. Define cone and cylinder
2. Define right circular cone
3. Define right circular cylinder
4. Define enveloping cone of a sphere
5. find the equation of a right circular cylinder whose axis is z axis and vertex origin.
6. Find the enveloping cone of a sphere $x^2+y^2+z^2+2x-2y-2=0$ with the vertex at $(1,1,1)$
7. Find the enveloping cone of a sphere $x^2+y^2+z^2-2x+4z=0$ with the vertex at $(1,1,1)$
1. Find the equation of the cone whose vertex is $(1,1,0)$ and whose guiding curve is $y=0; x^2+z^2=4$
2. Find the equation of the cone whose vertex is $(1,1,3)$ and whose guiding curve is $y=4; 4x^2+z^2=1$
3. Find the equation of the cone whose vertex is $(1,1,1)$ and whose guiding curve is $z=0; x^2+z^2=4$
4. Find the equation of the right circular cone whose vertex is $P(2,-3,5)$ axis PQ which makes equal angles with the axes and semi vertical angle 30°
5. Find the equation of a right circular cone whose vertex is $(2,-3,5)$ axis PQ which makes equal angles with the axes and which pass through $A(1,-2,3)$

6. Find the equation of cylinder whose generators are parallel to the line $x/1=y/-2=z/3$ and whose guiding curve is the ellipse $x^2+2y^2=1; z=3$
7. Find the equation of cylinder whose generators are parallel to the line $x/1=y/-2=z/3$ and whose guiding curve is $x^2+2y^2=1; z=0$
8. Find the equation of the right circular cylinder whose generators are parallel to $x/1=y/2=z/3$ and guiding curve $x^2+y^2=16; z=0$
9. Find the equation of the right circular cylinder of radius 2 whose axis passes through point $(1,2,3)$ and has direction cosines proportional to $(2,-3,6)$
10. Find the equation of right circular cylinder whose axis is $(x-1)/2 = y/3 = (z-3)/1$ and radius 2
11. Find the equation of the right circular cylinder of radius 3 and having for its axis the line $(x-1)/2 = (y-3)/2 = (5-z)/7$
12. Find the equation of the right circular cylinder of radius 3 whose axis passes through $(1,-1,2)$ and has direction cosines proportional to $1,2,2$